

Bill of Materials — T4R1FLASH

Component	Mass (g/cell)	kg/kWh	Caliber / source
Positive electrode active material (NMC811)	12.565	0.8951	94 wt% of coating (lit. default, annotated)
Positive electrode conductive additive (C-black)	0.401	0.0286	3 wt% of coating (lit. default, annotated)
Positive electrode binder (PVDF)	0.401	0.0286	3 wt% of coating (lit. default, annotated)
Negative electrode active material (graphite)	8.332	0.5935	96 wt% of coating (lit. default, annotated)
Negative electrode conductive additive (C-black)	0.087	0.0062	1 wt% of coating (lit. default, annotated)
Negative electrode binder (CMC/SBR)	0.260	0.0185	3 wt% of coating (lit. default, annotated)
Separator (PE/PP, porous)	0.173	0.0123	parameter set: sep thickness x (1-porosity) x density x area
Electrolyte (EC/EMC + LiPF6, enhanced transport)	5.036	0.3587	pore volume x 1.2 g/cm3 (lit.) x fill 1.0; EXCLUDED from contract ED caliber
Positive current collector (Al)	2.773	0.1975	parameter set formula
Negative current collector (Cu)	5.521	0.3933	parameter set formula
TOTAL (model, excl. electrolyte)	30.513	2.1736	calc-energy mass_kg
TOTAL incl. electrolyte	35.548	2.5323	pore vol x 1.2 g/cm3 (lit.)
Cell energy	14.0377		$\int V \cdot I \, dt$ 1C discharge
Enclosure / tabs	Not modeled		no parameter in set